Has the smooth 23×80 model been done before?

A review of prior work on the parametrisation of $d\sigma/dm_{hh}$ in HEFT, EFT morphing, machine-learned surrogates, and smooth coefficient functions

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Abstract

Our primary model writes $d\sigma/dm_{hh} = \sum_{i=1}^{23} C_i(\theta)\, a_i(m_{hh})$, with each coefficient function a_i a penalised cubic spline times a threshold factor, fitted to the *bin integrals* of 3,989 POWHEG spectra with the exact Poisson variances. The smoothness of each a_i is set by the evidence, the posterior is exactly Gaussian, and a discrepancy term grows with the degree of cancellation. Because the model is so simple, we asked whether it has been done before. We reviewed the di-Higgs EFT literature, ATLAS and CMS morphing and reweighting, machine-learned surrogates, and the statistics literature. For each work we record what was done, how the coefficients were obtained, and what it means for us. **Every ingredient has a precedent; we found no instance of the combination.** The 23-term form and per-bin coefficients are the community standard; the per-bin weighted least-squares fit is, in substance, our free 23×80 fit. In statistics, our model is a *varying-coefficient model* estimated as a *penalised composite link model*. We found no work that makes the HEFT coefficients smooth functions of m_{hh}, fits them to bin integrals, or attaches a cancellation-dependent model error. We recommend presenting the model as established statistics applied to a problem where the field still uses per-bin tables, and offering the smooth $a_i(m)$ as a replacement for those tables.

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1 Our model, for reference

So that each prior work can be compared term by term, we restate the model.

The smooth 23×80 model

$$\frac{d\sigma}{dm}(m;\theta) = \sum_{i=1}^{23} C_i(\theta)\, a_i(m), \qquad a_i(m) = \phi(m) \sum_{j=1}^{40} \beta_{ij}\, B_j(m), \qquad \phi(m) = \sqrt{1 - 4m_h^2/m^2}, \quad (1)$$

$$\sigma_k(\theta) = \int_{e_k}^{e_{k+1}} \frac{d\sigma}{dm}\, dm = \sum_{i,j} C_i(\theta)\, \beta_{ij}\, I_{jk}, \qquad I_{jk} = \int_{e_k}^{e_{k+1}} \phi\, B_j\, dm. \quad (2)$$

- $C_i(\theta)$: the 23 monomials in $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$ of the NLO HEFT cross section (eq. (3)). Known.
- B_j: 40 cubic B-splines on 250–1500 GeV, knots uniform in $\sqrt{m - 250}$.
- Likelihood: Gaussian, with the Poisson variance $q \max(f, q)$ of the model prediction f, iterated (IRLS) to remove the Neyman-χ^2 bias; q is each spectrum's weight per event, recovered from the integer counts.
- Prior: second differences of each β_i. Gaussian with precision λ_i; the 23 λ_i maximise the evidence.
- Posterior: exactly Gaussian in the 920 β_{ij}; $\gamma \approx 602$ effective parameters.
- Model error: $\varepsilon(m) \sum_i |C_i a_i|$ added in quadrature, with $\ln \varepsilon$ quadratic in m; pull s.d. 1.00 on 1,000 held-out spectra.

The *free* 23×80 fit is the limit with one box function per bin and no prior: 23 independent numbers per bin.

The question for this review is therefore four questions:

(Q1) Has anyone fitted the 23 coefficients *per bin* from many runs? (the free fit)

(Q2) Has anyone made them *smooth functions* of m_{hh}?

(Q3) Has anyone fitted smooth functions to *bin integrals* of counts, with the smoothness chosen from the data?

(Q4) Has anyone modelled the *error of the polynomial form itself*, and found it related to cancellation?

2 How we searched

We ran web searches (October 2026) on arXiv, journal sites, CERN and Fermilab Indico, and the statistics literature. We then read the method sections of the key papers, and the appendices where the coefficients are defined. The searches covered: the NLO HEFT $gg \to hh$ coefficients and how they are obtained; HH benchmark and reweighting papers; EFT morphing in ATLAS and CMS; "EFT-aware" histograms; neural surrogates and simulation-based inference for di-Higgs; splines and Gaussian processes for EFT predictions; spline fits to histograms; composite link models; and varying-coefficient models.

Limits of the search

A web search is not exhaustive. Experiment-internal notes are not public, and a smooth fit could be buried in a thesis or in the ancillary files of a paper. Before submission we should (i) run an INSPIRE "cited by" search on refs. [1, 2, 4], (ii) read the HH reweighting talks at the LHC Higgs WG (e.g. the 2025 WG4 meeting on "HH signal reweighting for EFT interpretations"), and (iii) ask the HH subgroup conveners directly whether anyone has tried a continuous m_{hh} parametrisation.

3 Di-Higgs in HEFT: the 23-term form and its coefficients

3.1 Buchalla, Capozi, Celis, Heinrich, Scyboz (2018) [1]

What they did. First NLO QCD calculation of $gg \to hh$ in the non-linear EFT (HEFT) with full top-mass dependence. They showed that the NLO cross section is a polynomial with 23 terms in the five couplings:

$$\begin{aligned}\sigma^{\mathrm{NLO}} = {} & A_1 c_t^4 + A_2 c_{tt}^2 + A_3 c_t^2 c_{hhh}^2 + A_4 c_{ggh}^2 c_{hhh}^2 + A_5 c_{gghh}^2 + A_6 c_{tt} c_t^2 + A_7 c_t^3 c_{hhh} \\ & + A_8 c_{tt} c_t c_{hhh} + A_9 c_{tt} c_{ggh} c_{hhh} + A_{10} c_{tt} c_{gghh} + A_{11} c_t^2 c_{ggh} c_{hhh} + A_{12} c_t^2 c_{gghh} \\ & + A_{13} c_t c_{hhh}^2 c_{ggh} + A_{14} c_t c_{hhh} c_{gghh} + A_{15} c_{ggh} c_{hhh} c_{gghh} \qquad \text{(the 15 LO terms)} \\ & + A_{16} c_t^3 c_{ggh} + A_{17} c_t c_{tt} c_{ggh} + A_{18} c_t c_{ggh}^2 c_{hhh} + A_{19} c_t c_{ggh} c_{gghh} + A_{20} c_t^2 c_{ggh}^2 \\ & + A_{21} c_{tt} c_{ggh}^2 + A_{22} c_{ggh}^3 c_{hhh} + A_{23} c_{ggh}^2 c_{gghh}. \end{aligned} \tag{3}$$

The first 15 terms are the LO polynomial; the eight terms A_{16}–A_{23}, all containing c_{ggh}, first appear at NLO.

How the coefficients were obtained. For the total cross section the A_i were obtained "via projections and performing a fit", and the two agree; only the projections are tabulated, with errors propagated from the input cross sections (correlations neglected). For $d\sigma/dm_{hh}$ (their appendix A.1): "The coefficients have been determined by evaluating 23 differential cross sections, obtained by changing the values of the five coupling parameters. Then the system of 23 equations is solved to extract the value of each coefficient." Bins are 20 GeV wide, 40 bins with centres 250–1030 GeV, in fb/GeV, published as ancillary CSV files.

What it means for us.

- Equation (3) is our $\sum_i C_i(\theta) a_i$; the C_i are these monomials. Our ordering and normalisation should be stated relative to theirs in the paper.
- Their differential coefficients come from a *square* 23×23 solve per bin: exact interpolation through 23 runs, with no redundancy. Any noise in the 23 runs goes straight into the A_i, amplified by the condition number of the coupling matrix. Our fit uses 3,989 runs, so it is heavily over-determined and the noise averages down.
- The m_{hh} dependence is piecewise constant in 20 GeV bins; nothing is smooth.
- **(Q1)** partly; **(Q2)–(Q4)** no.

3.2 Capozi and Heinrich (2020) [3]

What they did. A shape analysis of the m_{hh} distribution in HEFT: scans of the coupling space, clustering of shapes, and an autoencoder trained on 10^5 distributions.

How the coefficients were obtained. Taken from ref. [1]: "evaluated in bins of width 20 GeV from 250 GeV to 1050 GeV, i.e. for 40 bins", provided as CSV tables in fb/GeV for 13, 14 and 27 TeV. They report: "The median of the statistical uncertainties of the differential coefficients" is at most 3%, but above $\sim$650 GeV "some A_i coefficients have uncertainties in the 20–30% range."

What it means for us. This is direct evidence that the per-bin tables are noisy in the tail. Our smooth a_i share information across bins, and our posterior s.d. is 2.9% of the POWHEG noise. A side-by-side plot of our $a_i(m)$ with their tabulated $\tilde{A}_i$ at 13 TeV would make the point well (section 8). **(Q2)–(Q4)** no.

3.3 Heinrich, Jones, Kerner, Scyboz (2020) [2]

What they did. Implemented the NLO HEFT calculation in the POWHEG-BOX: `ggHH` with anomalous couplings. This is the generator of our 3,989 spectra.

How the coefficients enter. The two-loop virtual amplitude is too slow to evaluate on the fly, so it is "computed via a grid which encodes the dependence of the virtual two-loop amplitude on the kinematic invariants" $\hat{s}, \hat{t}$. Grids are made for "23 linearly independent sets of couplings" (6,715 points), and the code derives "for each phase-space point, the coefficients $a_1, \ldots, a_{23}$ by interpolation". Once the user chooses the couplings, "the 23 grids are combined into one" in the first POWHEG stage. The Born and real-emission parts are evaluated exactly. They note that these a_i are those of the Born–virtual interference, not the A_i of the cross section. The grid uncertainty is "below 2% throughout the whole m_{hh} range, except for the first bin" (6%, and 12% for benchmark 5), because the grid points were generated with SM couplings and the low m_{hh} region is sparsely populated.

What it means for us.

- They do interpolate smoothly, but in $(\hat{s}, \hat{t})$ and for the amplitude, not for the m_{hh} spectrum. Different object, different purpose.

- **A possible source of our model error.** Our spectra are generated with a *combined* grid, the real emission is computed exactly, and the POWHEG Sudakov factor reweights the Born. The 23-term form holds exactly at fixed order, but each of these steps can introduce small, coupling-dependent departures from it. Their 6–12% first-bin grid error is consistent with our larger residuals near threshold (deviance per bin 1.57 below 555 GeV).
- **(Q4)** indirectly: they quantify the grid's own error, not departures from the polynomial.

3.4 LHC Higgs WG note: Alasfar et al. (2023) [4]

What they did. The community reference for EFT descriptions of HH production, in HEFT and SMEFT, with benchmark points, translation between the two, and a reweighting recipe for the experiments.

How the coefficients were obtained. With POWHEG-BOX-V2 `ggHH` at NLO, 13 TeV, PDF4LHC15, $\mu_R = \mu_F = c\, m_{hh}/2$ with $c \in \{\frac{1}{2}, 1, 2\}$. The inclusive $\mathbf{A}$ and differential $d\mathbf{A}$ coefficients are obtained "using a weighted least square fit" to 63 simulations (62 HEFT points and the SM), following ref. [1]. The note defines

$$\frac{d\sigma}{dm_{hh}} = \mathrm{Poly}(\boldsymbol{c}, d\mathbf{A} \,|\, m_{hh}) = \boldsymbol{c}^{\mathsf{T}} \cdot d\mathbf{A}, \qquad w_{\mathrm{HEFT}} = \frac{\mathrm{Poly}(\boldsymbol{c}, d\mathbf{A} \,|\, m_{hh})}{\mathrm{Poly}(\boldsymbol{c}_{\mathrm{SM}}, d\mathbf{A} \,|\, m_{hh})}, \tag{4}$$

with bins of 20 GeV from 250 to 1050 GeV, then 1050–1200 and 1200–1400 GeV. Covariance matrices for $\mathbf{A}$ and $d\mathbf{A}$ are provided, with error-propagation formulas, and three coefficient sets for the scale variations.

Caveats they state.

- Events above 1400 GeV take the weight of the last bin: "not exactly correct, but it is the best available estimate."
- "The total cross section is better predicted by the inclusive $\mathbf{A}$ coefficients", so the inclusive set should fix the normalisation.
- The m_{hh} shapes close well; $p_T(h)$ shows discrepancies up to ∼40% (benchmark 1) and 10–20% elsewhere, because only m_{hh} is used.

What it means for us.

- **(Q1) yes.** This is our free 23×80 fit in substance: per-bin weighted least squares over many POWHEG runs, with a per-bin covariance. We must cite it as the baseline. Differences: we have 3,989 runs (63 times more), 15 GeV bins to 1455 GeV, the exact Poisson variance, and IRLS rather than weights from the observed counts. (The note does not define its weights; if they are the observed variances, the fit carries the Neyman bias we found and corrected, section 6.4.)
- Their need for separate inclusive coefficients is a symptom of the binned representation. A density model integrates to the total cross section at no extra cost.
- The "last bin" rule above 1400 GeV and the coarse last two bins are exactly the problems a smooth density solves inside its range. (Our own model should still not be used beyond 1455 GeV.)
- **(Q2)–(Q4)** no.

3.5 Cadamuro, Ingebretsen Carlson, Sjölin (2025) [8]

What they did. Extended the WG reweighting to several kinematic variables at 13 and 13.6 TeV, in HEFT and SMEFT, combining several reference samples with a convex combination weighted by a distance measure.

How the coefficients were obtained. "The coefficients of the polynomials, **A**, are derived by performing a fit to MC simulations at parton level": 63 samples for HEFT and 70 for SMEFT, couplings drawn uniformly in $[-4\pi, 4\pi]$. Again one polynomial per bin.

Key findings. Reweighting in m_{hh} alone reproduces p_T^h for constructive interference but not for destructive interference, with tail discrepancies of $\sim$20%: they identify destructive interference as "the main source for the discrepancies". They also note that POWHEG's Sudakov form factor makes the coupling dependence of variables affected by jet radiation deviate from a strict polynomial.

What it means for us. This is the closest thing we found to **(Q4)**. Their observable and question differ (a second variable, not the m_{hh} fit itself), but the physics is the same: where the 23 terms cancel, small departures from the polynomial become large relative errors. Our discrepancy $\varepsilon(m) \sum_i |C_i a_i|$ is a quantitative model of exactly that, and their Sudakov remark offers a mechanism. We should cite it when introducing the discrepancy.

3.6 Carvalho et al. (2016, 2017) [5, 6, 7]

What they did. At LO (15 terms): cluster analysis of m_{hh} and $|\cos\theta^*|$ shapes to choose the 12 benchmark points [5]; an analytical parametrisation of the total cross section [6]; and a per-bin parametrisation for reinterpreting searches [7].

How the coefficients were obtained.

- Total cross section [6]: "The components of **A** are extracted by maximizing the likelihood simultaneously for all the coefficients", with MINUIT, from 169–266 MadGraph points per energy (10,000 events per point, different seeds). The per-point error is set *a posteriori* to 0.03%, chosen so that the residuals have RMS ≈ 1.
- Differential [7]: 59 bins in m_{hh} (10 GeV up to 700 GeV, coarser above) by 4 in $|\cos\theta^*|$, one coefficient set per bin, from a simultaneous fit with MC statistical errors only (50,000 events per point, 13 million for the SM). They note that with only 15 samples per bin the result "would have a limited reliability", and use lower-dimensional sub-spaces to keep enough events per bin.

What it means for us. The over-determined per-bin fit, **(Q1)**, goes back to 2017 at LO. Setting the error a posteriori so that residuals have unit RMS is a crude form of what our discrepancy term does properly. **(Q2), (Q3)** no.

Work	Order	m_{hh} dependence	How coefficients are obtained
Carvalho 2017 [7]	LO, 15	59 $m_{hh} \times 4 \cos\theta^*$ bins	simultaneous fit per bin; 50k events/point
Buchalla 2018 [1]	NLO, 23	40 bins of 20 GeV	square 23×23 solve per bin from 23 runs
Capozi 2020 [3]	NLO, 23	as above	from [1]; 20–30% errors above 650 GeV
Heinrich 2020 [2]	NLO	per phase-space point	23 amplitude grids in $(\hat{s}, \hat{t})$, combined linearly
Alasfar 2023 [4]	NLO, 23	20 GeV to 1050, then 2 wide bins	weighted least squares, 63 runs; covariances
Cadamuro 2025 [8]	NLO, 23	per bin, several variables	fit to 63 (HEFT) / 70 (SMEFT) runs
This work	NLO, 23	**smooth** $a_i(m)$, 250–1455 GeV	penalised Poisson fit to bin integrals of 3,989 runs; evidence; exact posterior

Table 1: How the coefficient functions are represented in the di-Higgs EFT literature. Before this work the m_{hh} dependence is always piecewise constant.

4 Morphing and reweighting in the experiments

4.1 ATLAS morphing [9] and `RooLagrangianMorphFunc` [10]

What it is. Any distribution at couplings θ is written as a linear combination of N base distributions,

$$T(\theta) = \sum_{n=1}^{N} w_n(\theta)\, T(\theta_n), \tag{5}$$

where N equals the number of terms in the polynomial and the weights $w_n(\theta)$ follow from inverting the square matrix of monomials at the base points θ_n. It is implemented in ROOT as `RooLagrangianMorphFunc`.

What it means for us. Same algebra as eq. (1): the a_i are linear combinations of base histograms. Morphing applies it to whole histograms, with exactly as many samples as unknowns, so it is an interpolation, not a fit. There is no smoothing in the observable, and the statistical errors of distant points are amplified by large weights. **(Q1)** no (square system); **(Q2)–(Q4)** no.

4.2 Overdetermined morphing: Belyaev, Konoplich, Prokofiev (2025) [11]

What they did. Use more base samples than unknowns and solve by least squares via a QR factorisation, $x = R^{-1}Q^{\mathsf{T}}y$, to stabilise morphing far from the base points. They note that standard morphing computes its coefficients without accounting for the uncertainties of the simulated cross sections; statistical power is tracked by the effective-weight and Kish ratios.

What it means for us. This is the morphing community arriving at the over-determined fit, **(Q1)**, in 2025. Again no smoothing across bins, and the least squares is unweighted. Our IRLS with Poisson variances is the statistically correct version of their step.

4.3 EFT-aware histograms in CMS [12]

What it is. In the CMS top-quark EFT analyses (`topcoffea`, `histEFT`), each simulated event carries a quadratic in the Wilson coefficients, fixed by MadGraph reweighting at enough points and solved with least squares. "Instead of storing the sum of weight values, EFT histograms store the sum of the weight parameterizations."

What it means for us. An event-level version of the per-bin polynomial; exact for the generated sample, piecewise constant in the observable. No smoothing; no model error. Not directly applicable to POWHEG NLO, where per-event reweighting to arbitrary couplings is not available in the same way.

5 Machine learning

5.1 Quadratic classifier: Chen, Glioti, Panico, Wulzer (2021) [13]

What they did. For one Wilson coefficient c, the ratio of EFT to SM differential cross sections is a quadratic in c at every point x of phase space. They build this in:

$$\hat{r}(x,c) = [1 + c\, n_\alpha(x)]^2 + [c\, n_\beta(x)]^2, \qquad f(x,c) = \frac{1}{1+\hat{r}(x,c)}, \tag{6}$$

with networks n_α, n_β trained as a classifier on EFT and SM events at several values of c. "The quadratic dependence of the differential cross section on the EFT Wilson coefficients is built-in." They contrast it with the parametrised classifier, which takes c as an input and does not enforce the form.

What it means for us. This is the closest conceptual precedent for HEFTNet and for any $\sum_i C_i(\theta) a_i(x)$ model: a known polynomial in the couplings multiplying learned functions of the kinematics. The differences are the target (an unbinned likelihood ratio from events, versus binned cross sections), the representation (networks versus a linear spline basis), and the uncertainty (none quoted, versus an exact posterior). Our finding that, for m_{hh}, a 920-parameter linear model matches a network is a useful counterpoint. **(Q2)** yes in spirit (smooth functions of x with built-in coupling dependence), but not for HH, not fitted to bins, and without uncertainties.

5.2 MadMiner [14] and di-Higgs SBI [15]

What they did. MadMiner uses morphing-aware likelihood-ratio estimators: the matrix element is "a polynomial over chosen Wilson coefficients". Mastandrea, Nachman and Plehn apply it to di-Higgs in SMEFT with three coefficients, "a polynomial up to squared terms, resulting in 10 fit parameters" and 10 morphing basis points, plus a parametrised classifier in which c "is promoted to a feature". They note reweighting "works well when the new parameter point is close to the original one".

What it means for us. Unbinned inference, a different goal. Confirms that the polynomial structure is the standard way to carry coupling dependence; no smooth m_{hh} coefficients.

5.3 Learned loop amplitudes: Bahl, Braun, Heinrich, Plehn, Revelli (2026) [16]

What they did. Neural surrogates for the one- and two-loop $gg \to hh$ amplitudes with full top-mass dependence, with calibrated uncertainties, identification of poorly described phase-space regions, and adaptive training.

What it means for us. Complementary: they emulate the amplitude (in $\hat{s}, \hat{t}$) for the SM, not the m_{hh} spectrum across couplings. Their emphasis on calibrated surrogate uncertainties matches ours. A combination (EFT amplitude surrogates feeding our spectrum model) is a possible future direction.

Gaussian-process emulators. GP emulators are common in nuclear EFT (chiral EFT truncation errors) and cosmology, but we found none for the HH m_{hh} spectrum over the five HEFT couplings.

6 Statistics: the model has a name

6.1 Varying-coefficient models: Hastie and Tibshirani (1993) [17]

What it is. A regression in which the coefficients of known covariates are themselves smooth functions of another variable, the "effect modifier" r:

$$\eta = \beta_0(r) + \sum_i x_i \, \beta_i(r). \tag{7}$$

The $\beta_i(r)$ are estimated with smoothing splines or, later, P-splines [18, 19].

The match. Set $x_i = C_i(\theta)$, $\beta_i = a_i$, $r = m_{hh}$, and $\eta = d\sigma/dm$: eq. (1) *is* a varying-coefficient model with 23 covariates, no intercept, and an identity link. This answers **(Q2)** at the level of statistical structure: smooth coefficient functions of a covariate are textbook.

6.2 Penalised composite link model: Eilers (2007) [20], Rizzi, Gampe, Eilers (2015) [21]

What it is. Counts y_k are observed in coarse groups (age groups, income brackets). The underlying fine-grained distribution γ is smooth. The composite link model writes the expected counts as

$$\mu = \mathbf{C}\,\gamma, \qquad \gamma = \exp(\mathbf{B}\alpha), \qquad y_k \sim \text{Poisson}(\mu_k), \tag{8}$$

where the composition matrix $\mathbf{C}$ sums the fine grid into the observed groups and $\mathbf{B}$ is a B-spline basis. The penalised log-likelihood $\ell(\alpha) - \frac{\lambda}{2}\|D\alpha\|^2$ with a difference penalty D is maximised by IRLS, and λ is chosen by AIC or BIC. Rizzi et al. show it recovers smooth distributions from very coarse, open-ended groups; the R package `ungroup` implements it.

The match. Our I_{jk} in eq. (2) is the composition matrix applied to the spline basis: the model is defined as a density and compared with data through its bin integrals. Our merged-bin test (fit 30 GeV bins, predict the 15 GeV split) is exactly the "ungrouping" that PCLM was designed for. This answers **(Q3)**. Differences:

- **Link.** PCLM uses a log link so the density is positive. Our a_i change sign, so we use the identity link, and the model is linear in $\boldsymbol{\beta}$.

- **Covariates.** PCLM smooths one density. We smooth 23 functions, combined with known weights $C_i(\theta_s)$ that differ for each of the 3,989 spectra.
- **Smoothing parameters.** AIC/BIC over one λ, versus 23 λ_i by maximising the evidence.
- **Uncertainty.** Sandwich or Bayesian intervals for γ, versus the exact posterior of the linear model.

6.3 Older and related ideas

- **Histosplines** (Boneva, Kendall, Stefanov 1971) [22]: splines whose integrals over each histogram bin equal the bin content. The first appearance of the integral constraint, as interpolation rather than fitting.
- **Evidence and REML for smoothing parameters**: MacKay [23] for the Bayesian evidence; Wood [24] for the REML/marginal-likelihood methods that are now standard in generalised additive models (`mgcv`). Our 23 λ_i follow this route.
- **P-splines in HEP**: Whitehorn, van Santen, Lafebre [25] fit tensor-product penalised B-splines to IceCube photon tables of up to six dimensions, by penalised least squares with one λ per dimension (the paper does not choose them automatically), with non-negativity and monotonicity where needed. It shows that spline surrogates for large Monte Carlo tables are accepted practice in HEP.

6.4 Poisson versus Neyman χ^2: Baker and Cousins (1984) [26]

Weighting a fit by the *observed* counts biases it low; the likelihood with the *predicted* variance, or the equivalent IRLS, removes the bias. This is why our fit iterates with q $\max(f, q)$. It is textbook, but the HH reweighting literature does not say which weights its least-squares fits use.

7 Synthesis

What is new

Our model is a **varying-coefficient penalised composite link model**: the varying-coefficient structure supplies $\sum_i C_i(\theta) a_i(m)$, the composite link supplies the bin integrals. We found neither the combination in statistics nor any smooth parametrisation of the HEFT coefficients in physics. Specific to this work:

1. continuous $a_i(m_{hh})$ for the 23 NLO HEFT terms, with the threshold factor, fitted jointly to ∼4,000 POWHEG spectra;
2. a likelihood on the *integral* of the density over each bin, with the exact Poisson variances recovered from the integer event counts;
3. 23 smoothness strengths from the evidence and an exact Gaussian posterior for every $a_i(m)$ and every prediction;
4. a calibrated model error that scales with the cancellation $\sum_i |C_i a_i|$;
5. a test of within-bin accuracy that a per-bin table cannot pass by construction.

Ingredient	Closest prior work	Status
23-term NLO HEFT form	Buchalla et al. [1]	standard
Per-bin coefficients from many runs (free fit)	Carvalho [7], Alasfar [4], Belyaev [11]	standard
Smooth functions of x times known polynomial in couplings	Chen et al. [13]; VCM [17]	known, not for HH
Spline density fitted to bin integrals of counts	PCLM [20, 21]; histospline [22]	known, not in HEP EFT
Smoothing parameters by evidence	MacKay [23], Wood [24]	standard in statistics
Exact Poisson variances from integer counts; IRLS	Baker–Cousins [26]	standard idea; count recovery is ours
Threshold factor $\phi(m)$	phase-space β factor	physics; trivial
Model error $\propto$ cancellation	Cadamuro et al. [8] (qualitative)	**not found**
Within-bin validation (merged-bin test)	PCLM ungrouping	**not found in HEP**
All of the above, for $d\sigma/dm_{hh}$	—	**not found**

Table 2: Each ingredient of the smooth model and its closest precedent.

What a referee might say, and our answer

"This is just the WG per-bin fit with smoothing."
Yes, and that is the point: the per-bin fit is the floor for any 23-term model, and we reach it with half the parameters (a third of the effective ones), continuous predictions, and a demonstrated gain inside bins and in the tail.

"Why not a network?"
We tried; an ensemble of HEFTNets reaches the same floor, but its uncertainty requires an ensemble or a Laplace mixture, and it is unreliable between bin centres (spread up to 20–90 times larger below 400 GeV).

"The discrepancy term is ad hoc."
It has three parameters, is fitted on training spectra and calibrated on held-out ones (pull s.d. 1.00 in every cancellation fifth), and its form is motivated by the destructive-interference effect seen independently in ref. [8].

8 Recommendations and actions

For the paper.

1. In the introduction, cite [1, 2, 4] as the per-bin baseline and say that the free 23×80 fit reproduces that approach with a much larger sample.
2. Name the model: "a varying-coefficient model [17] estimated as a penalised composite link model [20, 21]".
3. Cite [13, 14] when motivating HEFTNet: polynomial in the couplings, learned functions of

the kinematics.

4. Cite [8] and [2] (grid, Sudakov) when introducing the cancellation-dependent discrepancy.
5. Offer the smooth $a_i(m)$ and their covariance as a drop-in replacement for the per-bin $d\mathbf{A}$ tables of [4]: same eq. (4), evaluated at any m_{hh} or integrated over any binning.

Concrete follow-ups.

1. Download the 13 TeV ancillary CSV of [1] (and the WG $d\mathbf{A}$ files if public) and overlay their $\tilde{A}_i$ on our $a_i(m)$ with bands, after matching conventions. Agreement validates both; differences in the tail show the gain.
2. Ask whether the WG fit weights are observed or predicted variances.
3. INSPIRE "cited by" search and a note to the HH conveners (see the box in section 2).
4. Merge the entries of `docs/smooth/literature_refs.bib` into `docs/paper/appendix_refs.bib`.

References

[1] G. Buchalla, M. Capozi, A. Celis, G. Heinrich, L. Scyboz, *Higgs boson pair production in non-linear Effective Field Theory with full m_t-dependence at NLO QCD*, JHEP 09 (2018) 057, arXiv:1806.05162.

[2] G. Heinrich, S. P. Jones, M. Kerner, L. Scyboz, *A non-linear EFT description of $gg \to HH$ at NLO interfaced to POWHEG*, JHEP 10 (2020) 021, arXiv:2006.16877.

[3] M. Capozi, G. Heinrich, *Exploring anomalous couplings in Higgs boson pair production through shape analysis*, JHEP 03 (2020) 091, arXiv:1908.08923.

[4] L. Alasfar et al., *Effective Field Theory descriptions of Higgs boson pair production*, LHC Higgs WG note, arXiv:2304.01968.

[5] A. Carvalho, M. Dall'Osso, T. Dorigo, F. Goertz, C. A. Gottardo, M. Tosi, *Higgs pair production: choosing benchmarks with cluster analysis*, JHEP 04 (2016) 126, arXiv:1507.02245.

[6] A. Carvalho et al., *Analytical parametrization and shape classification of anomalous HH production in the EFT approach*, LHCHXSWG-2016-001, arXiv:1608.06578.

[7] A. Carvalho, F. Goertz, K. Mimasu, M. Gouzevitch, A. Aggarwal, *On the reinterpretation of non-resonant searches for Higgs boson pairs*, arXiv:1710.08261.

[8] L. Cadamuro, T. Ingebretsen Carlson, J. Sjölin, *Di-Higgs and Effective Field Theory: Signal Reweighting Beyond m_{hh}*, arXiv:2502.20976.

[9] ATLAS Collaboration, *A morphing technique for signal modelling in a multidimensional space of coupling parameters*, ATL-PHYS-PUB-2015-047.

[10] ROOT, `RooLagrangianMorphFunc` class reference, https://root.cern.ch/doc/v632/classRooLagrangianMorphFunc.html.

[11] N. Belyaev, R. Konoplich, K. Prokofiev, *Signal Modelling with Overdetermined Morphing Technique*, arXiv:2512.12473.

[12] K. Mohrman, *EFT-aware histograms*, LPC EFT Workshop, Fermilab, 6 Sept. 2023, https://indico.fnal.gov/event/60025/.

[13] S. Chen, A. Glioti, G. Panico, A. Wulzer, *Parametrized classifiers for optimal EFT sensitivity*, JHEP 05 (2021) 247, arXiv:2007.10356.

[14] J. Brehmer, K. Cranmer, G. Louppe, J. Pavez, *Constraining Effective Field Theories with Machine Learning*, Phys. Rev. Lett. 121 (2018) 111801, arXiv:1805.00013.

[15] R. Mastandrea, B. Nachman, T. Plehn, *Constraining the Higgs Potential with Neural Simulation-based Inference for Di-Higgs Production*, arXiv:2405.15847.

[16] H. Bahl, J. Braun, G. Heinrich, T. Plehn, R. Revelli, *How to Trust Learned Loop Amplitudes*, arXiv:2601.00950.

[17] T. Hastie, R. Tibshirani, *Varying-coefficient models*, J. R. Stat. Soc. B 55 (1993) 757.

[18] P. H. C. Eilers, B. D. Marx, *Flexible smoothing with B-splines and penalties*, Statist. Sci. 11 (1996) 89.

[19] P. H. C. Eilers, B. D. Marx, *Practical Smoothing: The Joys of P-splines*, Cambridge University Press, 2021.

[20] P. H. C. Eilers, *Ill-posed problems with counts, the composite link model and penalized likelihood*, Statistical Modelling 7 (2007) 239.

[21] S. Rizzi, J. Gampe, P. H. C. Eilers, *Efficient estimation of smooth distributions from coarsely grouped data*, Am. J. Epidemiol. 182 (2015) 138.

[22] L. I. Boneva, D. Kendall, I. Stefanov, *Spline transformations: three new diagnostic aids for the statistical data-analyst*, J. R. Stat. Soc. B 33 (1971) 1.

[23] D. J. C. MacKay, *Bayesian interpolation*, Neural Comput. 4 (1992) 415.

[24] S. N. Wood, *Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models*, J. R. Stat. Soc. B 73 (2011) 3.

[25] N. Whitehorn, J. van Santen, S. Lafebre, *Penalized splines for smooth representation of high-dimensional Monte Carlo datasets*, Comput. Phys. Commun. 184 (2013) 2214, arXiv:1301.2184.

[26] S. Baker, R. D. Cousins, *Clarification of the use of chi-square and likelihood functions in fits to histograms*, Nucl. Instrum. Meth. 221 (1984) 437.